



SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:
Regulation (EC) No. 1907/2006 as amended by Regulation (EU) No. 2020/878, and
Regulation (EC) No. 1272/2008

Revision date 04-18-2025

Revision Number 4

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Product Code(s) 3751

Product Name 3751 ALMAGARD® VARI-PURPOSE LUBRICANT

Other means of identification

Mixture.

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended use Lubricant

Uses advised against Consumer use

1.3. Details of the supplier of the safety data sheet

Importer

HH Compliance Limited
Block 3 Harcourt Centre
Harcourt Road
Dublin 2
Ireland
Reg. Number: 421878
+353-21-4868122

Manufacturer Address

Lubrication Engineers, Inc.
1919 E. Tulsa
Wichita, KS 67216
USA
1-800-537-7683

For further information, please contact

Non-Emergency Telephone Number REACH +353-21-4868122
Other 1-800-537-7683

1.4. Emergency telephone number

Website www.llelubricants.com

Emergency Telephone CHEMTREC: 1-800-424-9300 (NORTH AMERICA)
+1-703-527-3887 (INTERNATIONAL)

Emergency Telephone - §45 - (EC)1272/2008

Europe 112

SECTION 2: Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Hazardous to the aquatic environment - chronic

Category 3 - (H412)

2.2. Label elements

Hazard statements

H412 - Harmful to aquatic life with long lasting effects.

EUH208 - Contains Calcium Alkarylsulfonate May produce an allergic reaction.

Precautionary Statements - EU (§28, 1272/2008)

P273 - Avoid release to the environment.

P501 - Dispose of contents and container in accordance with local, regional, national, and international regulations as applicable.

Unknown aquatic toxicity

Contains 88.345 % of components with unknown hazards to the aquatic environment.

Additional information

This product requires tactile warnings if supplied to the general public.

2.3. Other hazards

Other hazards Causes mild skin irritation.

PBT & vPvB The components in this formulation do not meet the criteria for classification as PBT or vPvB.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors.

SECTION 3: Composition/information on ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical name	Weight-%	REACH registration number	EC No (EU Index No)	Classification according to Regulation (EC) No. 1272/2008 [CLP]	Specific concentration limit (SCL)	M-Factor	M-Factor (long-term)	Notes
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	80 - 90	01-2119467170-45	265-155-0 (649-465-00-7)	Not classified (Note L)	-	-	-	L
antimony dialkyldithiocarbamate 15890-25-2	1 - 5	No data available	240-028-2 (051-003-00-9)	Acute Tox. 4 (H302) Acute Tox. 4 (H332) Aquatic Chronic 2 (H411) Aquatic Chronic 4 (H413)	-	-	-	A,1
dilithium azelate 38900-29-7	1 - 5	01-2120119814-57	254-184-4	Acute Tox. 4 (H302)	-	-	-	-
Calcium Alkarylsulfonate LE21112-01	0.1 - 1	No data available	-	Skin Sens. 1 (H317)	-	-	-	-
Lithium Hydroxide 1310-66-3	0.1 - 1	01-2119560576-31	215-183-4	Acute Tox. 3 (H301) Skin Corr. 1B (H314) Eye Dam. 1 (H318) Aquatic Acute 3 (H402) Acute Tox. 3 (H331)	-	-	-	-

				Acute Tox. 3 (H311)				
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Note A - Without prejudice to Article 17(2) of Regulation (EC) No 1272/2008, the name of the substance must appear on the label in the form of one of the designations given in Part 3 of Annex VI to that Regulation. In that Part, use is sometimes made of a general description such as "... compounds" or "... salts". In this case, the supplier who places such a substance on the market is required to state on the label the correct name, due account being taken of Section 1.1.1.4 of Annex VI to Regulation (EC) No 1272/2008.

Note L - The harmonized classification as a carcinogen applies unless it can be shown that the substance contains less than 3 % of dimethyl sulphoxide extract as measured by IP 346 ('Determination of polycyclic aromatics in unused lubricating base oils and asphaltene free petroleum fractions – Dimethyl sulphoxide extraction refractive index method' Institute of Petroleum, London), in which case a classification in accordance with Title II of this Regulation shall be performed also for that hazard class.

Note 1 - The concentration stated or, in the absence of such concentrations, the generic concentrations set out in this Regulation are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture.

Full text of H- and EUH-phrases: see section 16

Acute Toxicity Estimate

If LD50/LC50 data is not available or does not correspond to the classification category, then the appropriate conversion value from CLP Annex I, Table 3.1.2, is used to calculate the acute toxicity estimate (ATEmix) for classifying a mixture based on its components

Chemical name	Oral LD50 mg/kg	Dermal LD50 mg/kg	Inhalation LC50 - 4 hour - dust/mist - mg/L	Inhalation LC50 - 4 hour - vapor - mg/L	Inhalation LC50 - 4 hour - gas - ppm
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	5000 24000	5000	No data available	No data available	No data available
antimony dialkyldithiocarbamate 15890-25-2	16400	16000	No data available	No data available	No data available
Lithium Hydroxide 1310-66-3	120 210	No data available	0.96 3.4	No data available	No data available

This product does not contain candidate substances of very high concern at a concentration $\geq 0.1\%$ (Regulation (EC) No. 1907/2006 (REACH), Article 59).

SECTION 4: First aid measures

4.1. Description of first aid measures

Inhalation	Remove to fresh air.
Eye contact	Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.
Skin contact	Wash skin with soap and water. In the case of skin irritation or allergic reactions see a physician.
Ingestion	Rinse mouth.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms	Prolonged contact may cause redness and irritation.
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4.3. Indication of any immediate medical attention and special treatment needed

Note to physicians Treat symptomatically.

SECTION 5: Firefighting measures

5.1. Extinguishing media

Suitable Extinguishing Media Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media Do not scatter spilled material with high pressure water streams.

5.2. Special hazards arising from the substance or mixture

Specific hazards arising from the chemical Not easily combustible. Product may ignite and burn at temperatures exceeding the flash point.

5.3. Advice for firefighters

Special protective equipment and precautions for fire-fighters Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Personal precautions Ensure adequate ventilation.

For emergency responders Use personal protection recommended in Section 8.

6.2. Environmental precautions

Environmental precautions See Section 12 for additional Ecological Information.

6.3. Methods and material for containment and cleaning up

Methods for containment Prevent further leakage or spillage if safe to do so.

Methods for cleaning up Take up mechanically, placing in appropriate containers for disposal.

Prevention of secondary hazards Clean contaminated objects and areas thoroughly observing environmental regulations.

6.4. Reference to other sections

Reference to other sections See section 8 for more information. See section 13 for more information.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Advice on safe handling Ensure adequate ventilation.

General hygiene considerations Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Storage Conditions Keep container tightly closed in a dry and well-ventilated place.

Storage class (TRGS 510) Not determined.

7.3. Specific end use(s)

Risk Management Methods (RMM) The information required is contained in this Safety Data Sheet.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

Exposure Limits

Chemical name	European Union	Austria	Belgium	Bulgaria	Croatia
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	-	-	TWA: 5 mg/m ³ STEL: 10 mg/m ³	TWA: 5.0 mg/m ³	-
antimony dialkyldithiocarbamate 15890-25-2	-	TWA: 0.5 mg/m ³ STEL 1.5 mg/m ³	TWA: 0.5 mg/m ³	-	TWA: 0.5 mg/m ³
Lithium Hydroxide 1310-66-3	-	-	-	-	STEL: 1 mg/m ³
Chemical name	Cyprus	Czech Republic	Denmark	Estonia	Finland
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	-	TWA: 5 mg/m ³ Ceiling: 10 mg/m ³	TWA: 1 mg/m ³ STEL: 2 mg/m ³	-	TWA: 5 mg/m ³
antimony dialkyldithiocarbamate 15890-25-2	-	TWA: 0.5 mg/m ³ Ceiling: 1.5 mg/m ³	TWA: 0.5 mg/m ³ STEL: 1 mg/m ³ except Stibine	-	TWA: 0.5 mg/m ³
Chemical name	France	Germany TRGS	Germany DFG	Greece	Hungary
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	-	-	-	TWA: 5 mg/m ³	-
antimony dialkyldithiocarbamate 15890-25-2	TWA: 0.5 mg/m ³	-	-	TWA: 0.5 mg/m ³	-
Chemical name	Ireland	Italy MDLPS	Italy AIDII	Latvia	Lithuania
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	TWA: 5 ppm STEL: 15 ppm	-	TWA: 5 mg/m ³	TWA: 5 mg/m ³	TWA: 1 mg/m ³ STEL: 3 mg/m ³
antimony dialkyldithiocarbamate 15890-25-2	TWA: 0.5 mg/m ³ STEL: 1.5 mg/m ³	-	TWA: 0.5 mg/m ³	-	-
Lithium Hydroxide 1310-66-3	STEL: 1 mg/m ³	-	-	-	-
Chemical name	Luxembourg	Malta	Netherlands	Norway	Poland

Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	-	-	TWA: 5 mg/m ³	TWA: 1 mg/m ³ STEL: 3 mg/m ³	TWA: 5 mg/m ³ Sk*
antimony dialkyldithiocarbamate 15890-25-2	-	-	TWA: 0.5 mg/m ³	TWA: 0.5 mg/m ³ STEL: 1.5 mg/m ³	-
Chemical name	Portugal	Romania	Slovakia	Slovenia	Spain
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	TWA: 5 mg/m ³	TWA: 5 mg/m ³ STEL: 10 mg/m ³	TWA: 5 ppm TWA: 1 mg/m ³ TWA: 5 mg/m ³ Ceiling: 3 mg/m ³	-	TWA: 5 mg/m ³ STEL: 10 mg/m ³
antimony dialkyldithiocarbamate 15890-25-2	TWA: 0.5 mg/m ³	-	TWA: 0.5 mg/m ³	-	TWA: 0.5 mg/m ³
Chemical name	Sweden		Switzerland	United Kingdom	
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	NGV: 1 mg/m ³		-	-	
antimony dialkyldithiocarbamate 15890-25-2	NGV: 0.25 mg/m ³		-	TWA: 0.5 mg/m ³ STEL: 1.5 mg/m ³	
Lithium Hydroxide 1310-66-3	-		-	STEL: 1 mg/m ³	

Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

Derived No Effect Level (DNEL) - Workers

Chemical name	Oral	Dermal	Inhalation
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	-	0.97 mg/kg bw/day [4] [6]	2.73 mg/m ³ [4] [6] 5.58 mg/m ³ [5] [6]
Lithium 12-hydroxystearate 7620-77-1	-	41 mg/kg bw/day [4] [6] 0.172 mg/cm ² [5] [6]	-
dilithium azelate 38900-29-7	-	13.5 mg/kg bw/day [4] [6] 0.172 mg/cm ² [5] [6]	-

Notes

- [4] Systemic health effects.
[5] Local health effects.
[6] Long term.

Derived No Effect Level (DNEL) - General Public

Chemical name	Oral	Dermal	Inhalation
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	0.74 mg/kg bw/day [4] [6]	-	1.19 mg/m ³ [5] [6]
Lithium 12-hydroxystearate 7620-77-1	41 mg/kg bw/day [4] [6] 41 mg/kg bw/day [4] [7]	0.086 mg/cm ² [5] [6]	-
dilithium azelate 38900-29-7	13.5 mg/kg bw/day [4] [6] 27 mg/kg bw/day [4] [7]	0.023 mg/cm ² [5] [6]	-

Notes

[4]	Systemic health effects.
[5]	Local health effects.
[6]	Long term.
[7]	Short term.

Predicted No Effect Concentration (PNEC)

Chemical name	Freshwater	Freshwater (intermittent release)	Marine water	Marine water (intermittent release)	Air
dilithium azelate 38900-29-7	0.023 mg/L	0.23 mg/L	0.0023 mg/L	-	-

Chemical name	Freshwater sediment	Marine sediment	Sewage treatment	Soil	Food chain
Petroleum distillates, hydrotreated heavy naphthenic 64742-52-5	-	-	-	-	9.33 mg/kg food

8.2. Exposure controls

Engineering controls	Apply technical measures to comply with the occupational exposure limits.
Personal protective equipment	
Eye/face protection	Wear safety glasses with side shields (or goggles).
Hand protection	Wear suitable gloves.
Skin and body protection	Wear suitable protective clothing.
Respiratory protection	Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.
Thermal hazards	No information available.
Environmental exposure controls	No information available.

SECTION 9: Physical and chemical properties9.1. Information on basic physical and chemical properties

Physical state	Paste / Gel
Appearance	red
Odor	Characteristic.

<u>Property</u>	<u>Values</u>	<u>Method</u>
Melting point / freezing point	No data available	

Initial boiling point and boiling range	No data available	
Flammability	No data available	
Flammability Limit in Air		
Upper flammability or explosive limits	No data available	
Lower flammability or explosive limits	No data available	
Flash point	204 °C	ASTM D92
Autoignition temperature	No data available	
Decomposition temperature		
pH	No data available	
pH (as aqueous solution)	No data available	
Kinematic viscosity	144 cSt @ 40°C	ASTM D445
Dynamic viscosity	No data available	
Water solubility	No data available	
Solubility(ies)	No data available	
Partition coefficient	No data available	
Vapor pressure	No data available	
Relative density	No data available	
Bulk density	No data available	
Liquid Density	No data available	
Relative vapor density	No data available	
Particle characteristics		
Particle Size	No data available	
Particle Size Distribution	No data available	

9.2. Other information

9.2.1. Information with regard to physical hazard classes

No information available

9.2.2. Other safety characteristics

No information available

SECTION 10: Stability and reactivity

10.1. Reactivity

Reactivity No information available.

10.2. Chemical stability

Stability Stable under normal conditions.

Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

10.3. Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

10.4. Conditions to avoid

Conditions to avoid None known based on information supplied.

10.5. Incompatible materials

Incompatible materials Strong oxidizing agents.

10.6. Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

SECTION 11: Toxicological information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Information on likely routes of exposure

Inhalation	Inhalation of vapors in high concentration may cause irritation of respiratory system.
Eye contact	May cause slight irritation.
Skin contact	Causes mild skin irritation.
Ingestion	May be harmful if swallowed. May cause gastrointestinal discomfort if consumed in large amounts.

Symptoms related to the physical, chemical and toxicological characteristics

Symptoms Prolonged contact may cause redness and irritation.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Acute toxicity Based on available data, the classification criteria are not met.

Chemical name	Oral LD50	Dermal LD50	Inhalation LC50
Petroleum distillates, hydrotreated heavy naphthenic	> 5000 mg/kg (Rat) > 24 g/kg (Rat)	> 5000 mg/kg (Rabbit)	= 2062 ppm (Rat) 4 h
antimony dialkyldithiocarbamate	> 16400 mg/kg (Rat)	> 16000 mg/kg (Rabbit)	-
Lithium Hydroxide	= 120 mg/kg (Rat) = 210 mg/kg (Rat)	-	= 0.96 mg/L (Rat) 4 h > 3.4 mg/L (Rat) 4 h

Skin corrosion/irritation Classification based on data available for ingredients. Causes mild skin irritation.

Serious eye damage/eye irritation Based on available data, the classification criteria are not met.

Respiratory or skin sensitization Based on available data, the classification criteria are not met.

Germ cell mutagenicity Based on available data, the classification criteria are not met.

Carcinogenicity Based on available data, the classification criteria are not met.

DMSO Disclaimer Product containing mineral oil with less than 3% DMSO extract as measured by IP 346

Reproductive toxicity Based on available data, the classification criteria are not met.

STOT - single exposure Based on available data, the classification criteria are not met.

STOT - repeated exposure Based on available data, the classification criteria are not met.

Aspiration hazard Based on available data, the classification criteria are not met.

11.2. Information on other hazards

11.2.1. Endocrine disrupting properties

Endocrine disrupting properties Based on available data, the classification criteria are not met.

11.2.2. Other information

SECTION 12: Ecological information

12.1. Toxicity

Ecotoxicity Harmful to aquatic life with long lasting effects.

Unknown aquatic toxicity Contains 88.345 % of components with unknown hazards to the aquatic environment.

Chemical name	Algae/aquatic plants	Fish	Toxicity to microorganisms	Crustacea
Petroleum distillates, hydrotreated heavy naphthenic	-	LC50: >5000mg/L (96h, Oncorhynchus mykiss)	-	EC50: >1000mg/L (48h, Daphnia magna)

12.2. Persistence and degradability

Persistence and degradability No information available.

12.3. Bioaccumulative potential

Bioaccumulation

Chemical name	Partition coefficient
dilithium azelate	-3.3

12.4. Mobility in soil

Mobility in soil No information available.

12.5. Results of PBT and vPvB assessment

PBT and vPvB assessment Based on available data, the classification criteria are not met.

Chemical name	PBT and vPvB assessment
Petroleum distillates, hydrotreated heavy naphthenic	Not PBT/vPvB
dilithium azelate	Not PBT/vPvB

12.6. Endocrine disrupting properties

Endocrine disrupting properties Based on available data, the classification criteria are not met.

12.7. Other adverse effects

Other adverse effects No information available.

PMT or vPvM properties Based on available data, the classification criteria are not met.

SECTION 13: Disposal considerations

13.1. Waste treatment methods

Waste from residues/unused products	Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.
Contaminated packaging	Do not reuse empty containers.
Other information	According to the European Waste Catalogue, Waste Codes are not product specific, but application specific. Waste codes should be assigned by the user based on the application for which the product was used.

SECTION 14: Transport information

IATA

14.1. UN number or ID number	Not regulated
14.2. UN proper shipping name	Not regulated
14.3. Transport hazard class(es)	Not regulated
14.4. Packing group	Not regulated
14.5. Environmental hazards	No
14.6. Special precautions for user	
Special Provisions	None

IMDG

14.1. UN number or ID number	Not regulated
14.2. UN proper shipping name	Not regulated
14.3. Transport hazard class(es)	Not regulated
14.4. Packing group	Not regulated
14.5. Environmental hazards	No
14.6. Special precautions for user	
Special Provisions	None
14.7. Maritime transport in bulk according to IMO instruments	No information available

RID

14.1. UN number or ID number	Not regulated
14.2. UN proper shipping name	Not regulated
14.3. Transport hazard class(es)	Not regulated

14.4. Packing group	Not regulated
14.5. Environmental hazards	No
14.6. Special precautions for user	
Special Provisions	None

ADR

14.1. UN number or ID number	Not regulated
14.2. UN proper shipping name	Not regulated
14.3. Transport hazard class(es)	Not regulated
14.4. Packing group	Not regulated
14.5. Environmental hazards	No
14.6. Special precautions for user	
Special Provisions	None

ADN

14.1. UN number or ID number	Not regulated
14.2. UN proper shipping name	Not regulated
14.3. Transport hazard class(es)	No information available
14.4. Packing group	Not regulated
14.5. Environmental hazard	No
14.6. Special precautions for user	
Special Provisions	None

SECTION 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

National regulations

Germany

Water hazard class (WGK) slightly hazardous to water (WGK 1)

Chemical Prohibition Ordinance (ChemVerbotsV) Not applicable

TRGS 905 Not applicable

Switzerland

Ordinance on the Incentive Tax on Volatile Organic Compounds (OVOC) SR 814.018 Not applicable
WPO (GSchV) SR 814.201; WPA (GSchG) SR 814.20 Class B

European Union

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work.

Authorizations and/or restrictions on use:

This product does not contain substances subject to authorization (Regulation (EC) No. 1907/2006 (REACH), Annex XIV) This product does not contain substances subject to restriction (Regulation (EC) No. 1907/2006 (REACH), Annex XVII)

Persistent Organic Pollutants

Not applicable

Named dangerous substances per Seveso Directive (2012/18/EU)

Chemical name	Lower-tier requirements (tons)	Upper-tier requirements (tons)
Petroleum distillates, hydrotreated heavy naphthenic - 64742-52-5	-	25000

Ozone-depleting substances (ODS) Regulation (EU) 2024/590

Not applicable.

International Inventories

TSCA	Contact supplier for inventory compliance status
DSL/NDL	Contact supplier for inventory compliance status
EINECS/ELINCS	Contact supplier for inventory compliance status
ENCS	Contact supplier for inventory compliance status
IECSC	Contact supplier for inventory compliance status
KECL	Contact supplier for inventory compliance status
PICCS	Contact supplier for inventory compliance status
AIIC	Contact supplier for inventory compliance status
NZIoC	Contact supplier for inventory compliance status
TCSI	Contact supplier for inventory compliance status

- **TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory
- DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List
- EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances
- ENCS** - Japan Existing and New Chemical Substances
- IECSC** - China Inventory of Existing Chemical Substances
- KECL** - Korean Existing and Evaluated Chemical Substances
- PICCS** - Philippines Inventory of Chemicals and Chemical Substances
- AIIC** - Australian Inventory of Industrial Chemicals
- NZIoC** - New Zealand Inventory of Chemicals
- TCSI** - Taiwan Chemical Substance Inventory

15.2. Chemical safety assessment

Chemical Safety Report No information available

SECTION 16: Other information

Key or legend to abbreviations and acronyms used in the safety data sheet

Full text of any hazard and/or precautionary statements referred to under Sections 2-15

- H301 - Toxic if swallowed
- H302 - Harmful if swallowed
- H311 - Toxic in contact with skin
- H314 - Causes severe skin burns and eye damage
- H317 - May cause an allergic skin reaction
- H318 - Causes serious eye damage
- H331 - Toxic if inhaled
- H332 - Harmful if inhaled
- H402 - Harmful to aquatic life
- H411 - Toxic to aquatic life with long lasting effects
- H413 - May cause long lasting harmful effects to aquatic life
- P273 - Avoid release to the environment
- P501 - Dispose of contents and container in accordance with local, regional, national, and international regulations as applicable

Legend

ACGIH	American Conference of Governmental Industrial Hygienists
AIDII	Italian Association of Industrial Hygienists
ADN	Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe)
ADR	Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)
AIIC	Australian Inventory of Industrial Chemicals
ATE	Acute Toxicity Estimate
ASTM	American Society for the Testing of Materials
bar	Biological Reference Values for Chemical Compounds in the Work Area
BAT	Biological tolerance values for occupational exposure
BEL	Biological exposure limits
bw	Body weight
Ceiling	Maximum limit value
CLP	Classification, Labelling and Packaging Regulation; Regulation (EC) No 1272/2008
CMR	Carcinogen, Mutagen or Reproductive Toxicant
DFG	German Research Foundation
DOT	Department of Transportation (United States)
DSL	Domestic Substances List (Canada)
ECHA	European Chemicals Agency
EC Number	European Community number
EmS	Emergency Schedule
ENCS	Existing and New Chemical Substances (Japan)
EPA	Environmental Protection Agency
EWC	European Waste Codes
GHS	Globally Harmonized System
IARC	International Agency for Research on Cancer
IATA	International Air Transport Association
IBC	International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk
ICAO	International Civil Aviation Organization
IECSC	Inventory of Existing Chemical Substances in China
IMDG	International Maritime Dangerous Goods
IMO	International Maritime Organization
ISO	International Organization for Standardization
KECI	Korean Existing Chemicals Inventory
LC50	Lethal Concentration to 50% of a test population
LD50	Lethal Dose to 50% of a test population (Median Lethal Dose)
MAL	Measuring Technical Hygienic Air Needs
MARPOL	International Convention for the Prevention of Pollution from Ships
MDLPS	Ministry of Labor and Social Policy
n.o.s.	Not Otherwise Specified
NOAEC	No Observed Adverse Effect Concentration
NOAEL	No Observed Adverse Effect Level
NOELR	No Observable Effect Loading Rate
NZIoC	New Zealand Inventory of Chemicals
OECD	Organization for Economic Cooperation and Development
OEL	Occupational exposure limits
PBT	Persistent, Bioaccumulative and Toxic substance
PICCS	Philippines Inventory of Chemicals and Chemical Substances
PMT	Persistent, Mobile and Toxic
PPE	Personal protective equipment
QSAR	Quantitative Structure Activity Relationship
REACH	Registration, Evaluation, Authorization, and Restriction of Chemicals (REACH) Regulation (EC 1907/2006)

RID	Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe)
SADT	Self-Accelerating Decomposition Temperature
SAR	Structure-activity relationship
SDS	Safety Data Sheet
SL	Surface Limit
STEL	Short Term Exposure Limit
STOT RE	Specific target organ toxicity - Repeated exposure
STOT SE	Specific target organ toxicity - Single exposure
SVHC	Substance of very high concern
TCSI	Taiwan Chemical Substance Inventory
TDG	Transport of Dangerous Goods (Canada)
TRGS	Technical Rule for Hazardous Substances
TSCA	Toxic Substances Control Act (United States)
TWA	Time-Weighted Average
UN	United Nations
VOC	Volatile organic compounds
vPvB	Very Persistent and Very Bioaccumulative
vPvM	Very Persistent and Very Mobile
Sen+	Sensitizer
Sk*	Skin designation
**	Hazard Designation

Classification procedure	
Classification according to Regulation (EC) No. 1272/2008 [CLP]	Method Used
Acute oral toxicity	Calculation method
Acute dermal toxicity	Calculation method
Acute inhalation toxicity - gas	Calculation method
Acute inhalation toxicity - vapor	Calculation method
Acute inhalation toxicity - dust/mist	Calculation method
Skin corrosion/irritation	Calculation method
Serious eye damage/eye irritation	Calculation method
Respiratory sensitization	Calculation method
Skin sensitization	Calculation method
Mutagenicity	Calculation method
Carcinogenicity	Calculation method
Reproductive toxicity	Calculation method
STOT - single exposure	Calculation method
STOT - repeated exposure	Calculation method
Chronic aquatic toxicity	Calculation method
Acute aquatic toxicity	Calculation method
Aspiration hazard	Calculation method
Ozone	Calculation method

Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)
U.S. Environmental Protection Agency ChemView Database
European Food Safety Authority (EFSA)
European Chemicals Agency (ECHA) Committee for Risk Assessment (ECHA_RAC)
European Chemicals Agency (ECHA) (ECHA_API)
Environmental Protection Agency
Acute Exposure Guideline Level(s) (AEGL(s))
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act
U.S. Environmental Protection Agency High Production Volume Chemicals
Food Research Journal
Hazardous Substance Database
International Uniform Chemical Information Database (IUCLID)

National Institute of Technology and Evaluation (NITE)
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)
NIOSH (National Institute for Occupational Safety and Health)
National Library of Medicine's ChemID Plus (NLM CIP)
National Library of Medicine's PubMed database (NLM PUBMED)
U.S. National Toxicology Program (NTP)
New Zealand's Chemical Classification and Information Database (CCID)
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications
Organization for Economic Co-operation and Development High Production Volume Chemicals Program
Organization for Economic Co-operation and Development Screening Information Data Set
World Health Organization

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Disclaimer

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End of Safety Data Sheet